

A CLASS OF ELIMINATION SELECTION PROCEDURES BASED ON RANKS

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SUMMARY

A sequential elimination selection procedure is given for the problem of selecting that population having largest translation parameter among several populations whose distributions differ only by translations using the indifference zone formulation. The procedure is based on ranks and the common distribution is not assumed known. The asymptotic expected number of observations when the probability of correct selection tends to one is computed and comparisons with other procedures are discussed.

1. Introduction

Let $\Pi_1, \Pi_2, \dots, \Pi_k$, $k \geq 2$, denote k populations, the distribution function of Π_i being $F(x - \theta_i)$ where F is an unknown continuous df with square integrable density f and θ_i is an unknown location parameter for Π_i . This paper is concerned with the problem of constructing sequential elimination procedures for selection of the population having maximum θ_i .

More specifically, if the indexing is such that $\theta_1 \leq \theta_2 \leq \dots \leq \theta_k$, then we wish to construct procedures having the property

$$(1) \quad P(\text{CS}) \geq P^* \quad \text{if } \theta_k - \theta_{k-1} \geq \tau^*$$

where CS denotes the event of a correct selection and P^* and τ^* are given numbers with $1/k < P^* < 1$ and $\tau^* > 0$.

Fixed sample size procedures for this problem are discussed in Rizvi and Woodworth (1970) where further references can be found. Sequential procedures without elimination based on the one sample Hodges-Lehmann estimate and on the generalized Hodges-Lehman estimate are given in Geertsema (1972) and Swanepoel and Geertsema (1973). Their procedures satisfy (1) only asymptotically as $\tau^* \rightarrow 0$.

Here we propose procedures based on a class of one sample rank order statistics which both eliminate and satisfy (1) non-asymptotically:

Let $X_{i\alpha}$ denote the α -th observation from Π_i (all such observations being assumed independent random variables); put

$$(2) \quad Y_{ij\alpha} = X_{i\alpha} - X_{j\alpha}, \tau_{ij} = \theta_i - \theta_j$$

and define

$$(3) \quad \alpha(u) = \begin{cases} 0 & \text{for } u < 0 \\ 1 & \text{for } u \geq 0. \end{cases}$$

Next let

$$(4) \quad R_{ij\alpha n}(\tau_{ij}) = \sum_{\beta=1}^n \alpha(|Y_{ij\alpha} - \tau_{ij}| - |Y_{ij\beta} - \tau_{ij}|)$$

and

$$(5) \quad Z_{ijn}(\tau_{ij}) = n^{-1} \sum_{\alpha=1}^n \alpha(Y_{ij\alpha} - \tau_{ij}) J_n(R_{ij\alpha n}(\tau_{ij}) / (n+1))$$

where $J_n(u)$, $0 < u < 1$, is a rank score defined by Sen and Ghosh (1971), p. 190; we use their notation and results in this regard without further explanation.

Then our ranks procedure is defined as follows: Let $\{c_n\}$ be a sequence of positive real numbers, depending on P^* , to be specified further below. Begin at stage one with a single observation from each of the k populations and compute the $k(k-1)$ statistics $Z_{ij1}(-\tau^*)$; if for some i , $Z_{ij1}(-\tau^*) - \frac{1}{2}J_1 \geq c_1$, then Π_j is eliminated from further consideration. If no populations remain as a result of this rule, sampling terminates and a Π_i for which $\min_{j \neq i} Z_{ij1}(-\tau^*)$ is maximized, is declared "best"; if only one population remains, it is declared best and if more than one remains, we proceed to the next stage. At the n -th stage, a single observation is made from each of the remaining populations, the statistics $Z_{ijn}(-\tau^*)$ are computed for all remaining (ordered) pairs and if, for some remaining Π_i ,

$$(6) \quad Z_{ijn}(-\tau^*) - \frac{1}{2}J_n \geq c_n$$

then Π_j is eliminated and the procedure repeated.

In paragraph 2 we prove that this procedure stops a.s. under suitable conditions and we discuss the probability of a correct selection; in paragraph 3 we consider the asymptotic expected number of observations (AENO) and the asymptotic expected number of stages (AENS) as $P^* \rightarrow 1$; finally in paragraph 5 we consider special cases and make comparisons.

2. Stopping and the probability of CS

We now show that if

$$(7) \quad c_n \rightarrow 0 \text{ as } n \rightarrow \infty$$

and if condition (12) below is satisfied, then the procedure stops with probability one for all parameter configurations.

Suppose to the contrary that for some sample point ω (in the underlying probability space), the procedure does not stop; then there are at least two populations Π_i and Π_j , say, which are never eliminated. Now, either $\tau_{ij} \geq 0$ or $\tau_{ji} \geq 0$; consider first the case $\tau_{ij} \geq 0$. From Theorem 1 of Sen (1970) we get

$$(8) \quad Z_{ijn}(-\tau^*) \rightarrow m_{ij}^* \text{ a.s.}$$

where

$$(9) \quad m_{ij}^* = \int_0^\infty J(G(x - \tau^* - \tau_{ij}) \wedge G(-x - \tau^* - \tau_{ij})) dG(x - \tau^* - \tau_{ij})$$

with

$$(10) \quad G(x) = P(Y_{ij\alpha} - \tau_{ij} \leq x) = \int_{-\infty}^\infty F(x+y) dF(y).$$

Noting that $G(x) + G(-x) = 1$ and making the substitution $z = x - \tau^* - \tau_{ij}$, we get

$$(11) \quad m_{ij}^* = \int_{-\tau^* - \tau_{ij}}^\infty J(G(z) + G(z + 2\tau^* + 2\tau_{ij}) - 1) dG(z) \\ \geq \int_{-\tau^*}^\infty J(G(z) + G(z + 2\tau^*) - 1) dG(z).$$

Now generally, for $\tau^* > 0$ we have

$$(12) \quad \int_{-\tau^*}^\infty J(G(z) + G(z + 2\tau^*) - 1) dG(z) > \int_0^\infty J(2G(z) - 1) dG(z) = \frac{1}{2}\mu.$$

Hence, if this holds, then apart from ω in a null set, we have $Z_{ijn}(-\tau^*) \rightarrow m_{ij}^* > \frac{1}{2}\mu = \lim_{n \rightarrow \infty} (\frac{1}{2}J_n^- + c_n)$ so that for large enough n we would have $Z_{ijn}(-\tau^*) - \frac{1}{2}J_n^- \geq c_n$ and eliminate Π_j - thus getting a contradiction. The case $\tau_{ji} \geq 0$ is handled similarly.

Henceforth we assume (7) and (12) holds. We then derive a lower bound for $P(\text{CS})$. Now

$$1 - P(\text{CS}) \leq \sum_{i \neq k} P(\Pi_i \text{ eliminates } \Pi_k)$$

$$\leq \sum_{i \neq k} P(Z_{ikn}(-\tau^*) - \frac{1}{2}\bar{J}_n \geq c_n \text{ for some } n \geq 1).$$

Outside the indifference zone we have $\tau_{ik} = \theta_i - \theta_k \leq \theta_{k-1} - \theta_k \leq -\tau^*$; it is readily seen that $Z_{ijn}(\tau_{ij})$ is non-increasing in τ_{ij} for all i and j ; hence $Z_{ikn}(\tau_{ik}) \geq Z_{ikn}(-\tau^*)$ and

$$1 - P(\text{CS}) \leq \sum_{i \neq k} P(Z_{ikn}(\tau_{ik}) - \frac{1}{2}\bar{J}_n \geq c_n \text{ for some } n \geq 1) \\ \leq 1 - P^*$$

provided that, for each $i \neq k$, we have

$$(13) \quad P(Z_{ikn}(\tau_{ik}) - \frac{1}{2}\bar{J}_n \geq c_n \text{ for some } n \geq 1) \leq (1 - P^*)/(k-1).$$

Hence, when (13) holds, (1) is satisfied.

We now indicate an approach by means of which $\{c_n\}$ can be selected so as to make (13) valid.

The left-hand side of (13) is less than or equal to

$$\sum_{n=1}^{\infty} P(Z_{ikn}(\tau_{ik}) - \frac{1}{2}\bar{J}_n \geq c_n) \\ \leq \sum_{n=1}^{\infty} \exp(-a_n c_n) E \exp(a_n (Z_{ikn}(\tau_{ik}) - \frac{1}{2}\bar{J}_n))$$

for $a_n > 0$ (using Markov's inequality). Since $Y_{ik\alpha} - \tau_{ik}$ is distributed symmetric about 0, we get from Hájek (1969), p. 105,

$$E \exp(a(Z_{ikn}(\tau_{ik}) - \frac{1}{2}\bar{J}_n)) = \prod_{\alpha=1}^n \cosh(\frac{1}{2}a \alpha^{-1} J_n(\alpha/(n+1))) \\ \leq \exp(\sum_{\alpha=1}^n a^2 J_n^2(\alpha/(n+1))/8n^2) \\ = \exp(a^2 A_n^2/8n).$$

Now $\exp(-ac_n + a^2 A_n^2/8n)$ is minimized by taking $a = a_n = 4nc_n/A_n^2$, so that we then get

$$P(Z_{ikn}(\tau_{ik}) - \frac{1}{2}\bar{J}_n \geq c_n \text{ for some } n \geq 1) \\ \leq \sum_{n=1}^{\infty} \exp(-2nc_n^2/A_n^2)$$

and (13) will be satisfied when c_n is such that

$$(14) \quad \sum_{n=1}^{\infty} \exp(-2nc_n^2/A_n^2) \leq (1 - P^*)/(k-1).$$

One possible choice is

$$(15) \quad c_n = n^{-1/2} A_n [\log n + 1/2 \log(\pi^2(k-1)/6(1-P^*))]^{1/2}$$

and this will be used in what follows.

3. Asymptotic expected number of observations as $P^* \rightarrow 1$

In order to get an expression for the AENO as $P^* \rightarrow 1$; we apply Theorem 3 of Swanepoel and Venter (1975); we take $Y_{ijn} = (Z_{ijn}(-\tau^*) - 1/2 \bar{J}_n)/A_n$ and $\rho(n, P^*) = c_n/A_n = n^{-1/2} [\log n + 1/2 \log(\pi^2(k-1)/6(1-P^*))]^{1/2}$. It is then clear that conditions (I) and (II) hold with

$$(16) \quad m_{ij} = (m_{ij}^* - 1/2 \mu)/A$$

where m_{ij}^* is given by (9) or (11) above and $\mu = \int_0^1 J(u) du$, $A^2 = \int_0^1 J^2(u) du$. Also (III) is readily seen to hold with

$$(17) \quad g(P^*) = \log((k-1)/(1-P^*)) \text{ (or } g(P^*) = \log(\pi^2(k-1)/6(1-P^*)))$$

and

$$(18) \quad \ell(x) = 1/2x^2.$$

In order to verify condition (IV), we define a random variable R which majorizes the number of stages N as follows:

For each ordered pair (ij) such that $\tau_{ij} \geq 0$, we have $m_{ij}^* - 1/2 \mu > 0$; also $\bar{J}_n \rightarrow \mu$; we can find $\epsilon > 0$ such that for all $n \geq n_1(\epsilon)$ we have $m_{ij}^* - 1/2 \bar{J}_n > 2\epsilon$ for all pairs (ij) such that $\tau_{ij} \geq 0$. Define R_{ij} as the smallest integer $n_2 \geq n_1(\epsilon)$ such that for all $n \geq n_2$ we have $|Z_{ijn}(-\tau^*) - m_{ij}^*| < \epsilon$. Next let M be the smallest integer $n_3 \geq n_1(\epsilon)$ such that for all $n \geq n_3$ we have $c_n < \epsilon$. Then take $R = \max(M, R_{ij} : \tau_{ij} \geq 0)$. To see that $N \leq R$ a.s., suppose to the contrary that $N > R$; then at stage R there exists two (at least) populations Π_i and Π_j which have not yet been eliminated; now, either $\tau_{ij} \geq 0$ or $\tau_{ji} \geq 0$; in the first case, we get $Z_{ijR}(-\tau^*) \geq m_{ij}^* - \epsilon$, i.e.

$$Z_{ijR}(-\tau^*) - 1/2 \bar{J}_n \geq m_{ij}^* - 1/2 \bar{J}_n - \epsilon \geq 2\epsilon - \epsilon = \epsilon > c_R$$

so that Π_j would be eliminated by Π_i — a contradiction. Similarly for $\tau_{ji} \geq 0$.

To show uniform integrability of $\{R/g(P^*)\}$ it suffices to show this for $\{R_{ij}/g(P^*)\}$, which follows if we can show that

$$(19) \sum_n P_0 < P^* < 1 \sup P(R_{ij}/g(P^*) > n) < \infty$$

for some $P_0 < 1$. (Bickel & Yahav (1968)). Now, let $r = [ng(P^*)]$, then

$$\begin{aligned} P(R_{ij}/g(P^*) > n) &\leq P(R_{ij} > r) \\ &= P(|Z_{ij\ell}(-\tau^*) - m_{ij}^*| \geq \epsilon \text{ for some } \ell \geq r) \\ &\leq \sum_{\ell=r}^{\infty} P(|Z_{ij\ell}(-\tau^*) - m_{ij}^*| \geq \epsilon). \end{aligned}$$

Using the bound (2.7) of Theorem 2 of Sen (1970), (19) follows and therefore $\{R/g(P^*)\}$ is uniformly integrable. Further since R_{ij} does not depend on P^* while $M \rightarrow \infty$ as $P^* \rightarrow 1$, it follows that for P^* large enough, $R = M$. Moreover, it is easily seen that $M/g(P^*) \rightarrow A^2/2\epsilon^2$ as $P^* \rightarrow 1$; hence also $R/g(P^*) \rightarrow A^2/2\epsilon^2$ a.s. as $P^* \rightarrow 1$; condition (IV) therefore holds. Hence, if T is the total number of observations required, then

$$(20) \lim_{P^* \rightarrow 1} ET/g(P^*) = \sum_{i \neq k} \frac{1}{2} A^2 / (m_{ki}^* - \frac{1}{2}\mu)^2 + \frac{1}{2} A^2 / (m_{k,k-1}^* - \frac{1}{2}\mu)^2$$

and $\lim_{P^* \rightarrow 1} EN/g(P^*)$ is given by the last term on the right where N is the number of stages. If we define the function

$$(21) S_{F,J}(x) = \frac{1}{2} A^2 / \left(\int_{-x}^{\infty} J(G(z) + G(z + 2x) - 1) dG(z) - \frac{1}{2}\mu \right)^2$$

then we get

$$(22) \lim_{P^* \rightarrow 1} ET/g(P^*) = \sum_{i \neq k} S_{F,J}(\tau^* + \theta_k - \theta_i) + S_{F,J}(\tau^* + \theta_k - \theta_{k-1}).$$

In particular, outside the indifference zone $\theta_k - \theta_i \geq \tau^*$ so that then

$$(23) \lim_{P^* \rightarrow 1} ET/g(P^*) \leq k S_{F,J}(2\tau^*).$$

4. Remarks

4.1 Procedures based on sample means

In this paragraph, assume that

$$(24) \quad E|X_{i\alpha}| < \infty, EX_{i\alpha} = \theta_i \text{ for all } i,$$

put $\bar{X}_{in} = n^{-1} \sum_{\alpha=1}^n X_{i\alpha}$ and define

$$(25) \quad Z_{ijn}(\tau_{ij}) = \bar{X}_{in} - \bar{X}_{jn} - \tau_{ij}.$$

Replace $\bar{J}_n$ by 0 and define an elimination selection procedure as in paragraph 1 based on this choice of $Z_{ijn}(-\tau^*)$. This procedure will be referred to as the sample means procedure. If $c_n \rightarrow 0$, this procedure stops a.s.

If, in addition

$$(26) \quad E \exp(a(X_{i\alpha} - \theta_i)) \leq \exp(\frac{1}{2}a^2\sigma^2)$$

with σ^2 a known finite constant (usually $\sigma^2 = \text{Var}(X_{i\alpha})$), then we can repeat the earlier argument to find that (1) holds if we select

$$(27) \quad c_n = n^{-\frac{1}{2}} 2\sqrt{2}\sigma[\log n + \frac{1}{2}\log(\pi^2(k-1)/6(1-P^*))]^{\frac{1}{2}}.$$

The AENO for this procedure can again be determined: if T_1 denotes the total number of observations required to stop, then we get

$$(28) \quad P^* \lim_{P^* \rightarrow 1} ET_1/g(P^*) = \sum_{i \neq k} \ell_1(\tau^* + \theta_k - \theta_i) + \ell_1(\tau^* + \theta_k - \theta_{k-1})$$

where

$$(29) \quad \ell_1(x) = 4\sigma^2/x^2.$$

In Swanepoel and Venter (1975) we discussed a class of procedures for the problem of selecting the largest mean for populations satisfying (26); the best procedure within this class in terms of smallest AENO, has an AENO given by (28) and (29). Thus, although the upper bounds used in our arguments to establish (1) may seem crude it nevertheless yields good results at least in the asymptotic sense when $P^* \rightarrow 1$.

By using other upper bounds for the left-hand side of (13) we have been able to show that (1) is satisfied for choices of c_n of the form

$$(30) \quad c_n = n^{-\frac{1}{2}} D_n [\log \log n + dg(P^*)]^{\frac{1}{2}}$$

where $\{D_n\}$ and d are positive constants subject to certain restrictions; however, in

terms of AENO as $P^* \rightarrow 1$ no improvement over the above results is obtained in this way.

4.2 The boundary of the indifference zone

In this paragraph we assume that $\theta_k - \theta_i = \tau^*$ for $i=1,2,\dots,k-1$. For this parameter configuration an asymptotic relative efficiency of the ranks procedure compared to the means procedure is

$$(31) \lim_{P^* \rightarrow 1} ET_1/ET = \sigma^2/\tau^{*2} S_{F,J}(2\tau^*).$$

This is difficult to evaluate in general. For $\tau^* \rightarrow 0$ (31) tends to

$$(32) 32\sigma^2 C^2/A^2, C = \int_0^\infty J'(2G(z) - 1)g(z)^2 dz$$

where $g(z) = G'(z)$. (32) represents the usual Pitman efficiency of the one-sample rank test based on sample differences as compared to the t-test based on sample differences for testing equality of two population means with equal sample sizes. Well-known facts about such Pitman efficiencies apply; e.g. if $J(u) = \Phi^{-1}((1+u)/2)$, (32) becomes

$$(33) 2\sigma^2 \left(\int_{-\infty}^{\infty} g(z)^2 dz / \phi(\Phi^{-1}(G(z))) \right)^2$$

and if $2\sigma^2$ is the variance of G (as it will usually be), then this expression is ≥ 1 for all distributions G (i.e. for all F) with equality only if G is normal.

4.3 Comparison with the non-eliminating procedures of Geertsema (1972) and Swanepoel and Geertsema (1973)

If T_2 denotes the total number of observations required for the non-eliminating procedure of Swanepoel and Geertsema (1973), then it was shown that

$$(34) \lim_{P^* \rightarrow 1} ET_2/g(P^*) = kA^2/4\tau^{*2}B^2$$

where

$$(35) B = \int_0^\infty J'(2F(x) - 1)f(x)^2 dx.$$

Hence, outside the indifference zone $\theta_k - \theta_i \geq \tau^*$, we get

$$(36) \lim_{P^* \rightarrow 1} ET_2/ET \geq A^2/4\tau^{*2}B^2 S_{F,J}(2\tau^*).$$

Again this expression is difficult to evaluate; for $\tau^* \rightarrow 0$, (36) tends to

$$(37) \quad 8C^2/B^2$$

with C as above. For the special choice $J(u) = u$ (Wilcoxon case), we have

$$C = \int_0^\infty g(z)^2 dz, \quad B = \int_0^\infty f(x)^2 dx$$

and (37) can be evaluated for some choices of f , e.g.

$$\text{if } f(x) = \phi(x), \quad 8C^2/B^2 = 4$$

$$\text{if } f(x) = \frac{1}{2}e^{-|x|}, \quad 8C^2/B^2 = 25/8$$

$$\text{if } f(x) = 1/\pi(1+x^2), \quad 8C^2/B^2 = 2.$$

Also, for $J(u) = \Phi^{-1}((1+u)/2)$ and $f(x) = \phi(x)$ we get $8C^2/B^2 = 4$.

4.4 Concluding remarks

The rank procedure defined here has several attractive features:

- (a) it satisfies the probability requirement (1) exactly (not just asymptotically as in the case e.g. with the procedures of Swanepoel and Geertsema (1973) and Puri and Puri (1969);
- (b) somewhat awkward assumptions on parameter configurations such as those of Puri and Puri (1969) are not required here;
- (c) except for the regularity assumptions mentioned in paragraph 1, no assumptions are made about the form of the distribution function F , unlike the case in e.g. Swanepoel and Geertsema (1973) where symmetry of F is assumed, or in Puri and Puri (1969) where F is assumed to be known;
- (d) in terms of AENO as $P^* \rightarrow 1$ it seems to compare quite favourably with other procedures as far as the above (admittedly limited) comparisons show. The procedures of Swanepoel and Geertsema (1973), Puri and Puri (1969), etc. are non-eliminating so that this is to be expected.

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